

Two-Phase PageRank with Travel-Time Self-Loops

Traffic Prediction on the Salt Lake City Road Network

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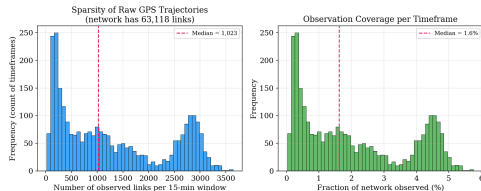
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Outline

- 1 Problem & Motivation
- 2 Building the Transition Matrix M_{2N}
- 3 Building the Teleportation Vector E_N
- 4 The PageRank Power Iteration
- 5 Smoothing Justification
- 6 Results
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The Problem

- Salt Lake City road network: **99,716 links**
- GPS data covers only a **tiny fraction** per 15-min window
- Goal: predict **full traffic distribution** from a sparse sample

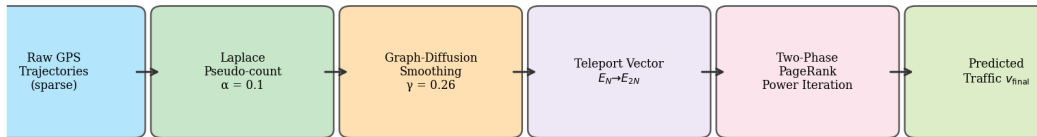


Key Idea

Model traffic as a **random walk** on the road graph $\Rightarrow$ PageRank

End-to-End Pipeline

Two-Phase PageRank: End-to-End Data Pipeline

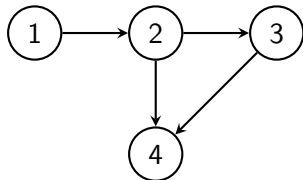


Step 1: Base Adjacency

- Directed graph: link $i \rightarrow j$ if road i connects to road j
- Row-stochastic normalisation:

$$P_{ij} = \frac{A_{ij}}{\sum_k A_{ik}}$$

- This is **purely topological** — no GPS data involved



$$P = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & \frac{1}{2} & \frac{1}{2} \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

Step 2: Travel-Time Self-Loops

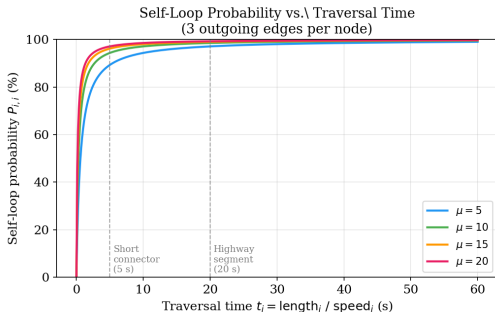
Problem: Standard PageRank treats every road transition as instantaneous.

Fix: Add a **self-loop** proportional to traversal time:

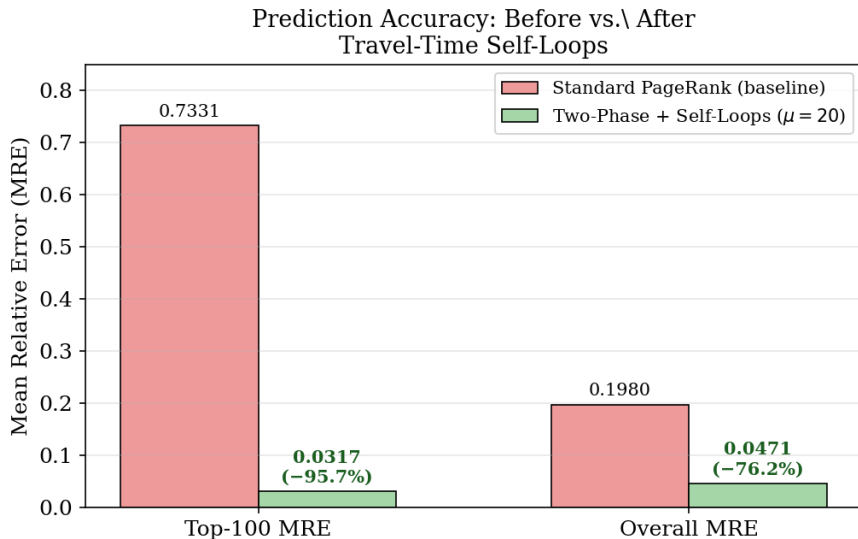
$$w_i = \mu \cdot \frac{\ell_i}{s_i}$$

- ℓ_i = road length, s_i = speed limit
- $\mu = 20$ = dwell-time scale

Longer roads trap more mass



Self-Loop: Before & After



Step 3: Two-Phase Block Matrix

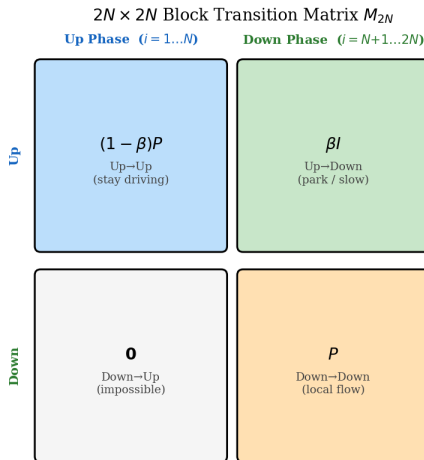
Split each link into two states:

- **Up** = actively driving through
- **Down** = locally dispersing

Block structure ($2N \times 2N$):

$$M_{2N} = \begin{pmatrix} (1 - \beta) P_{\text{self}} & \beta I \\ 0 & P_{\text{adj}} \end{pmatrix}$$

- $\beta = 0.90$: drop-rate to Down phase
- P_{self} : includes self-loops
- P_{adj} : local adjacency



Key Point: Transition Matrix Uses No Ground Truth

The transition matrix M_{2N} is built **entirely** from **road topology, lengths, and speed limits**.

No GPS data. No traffic counts. No ground truth.

Component	Data source	Role
Transition M_{2N}	Road network (static)	Where vehicles move
Teleportation E_{2N}	GPS data (dynamic)	Where vehicles are born

Four-Step Construction of E_N

- 1 **Extract raw GPS counts** $\mathbf{c} \in \mathbb{Z}_{\geq 0}^N$ (most $c_i = 0$)
- 2 **Laplace pseudo-count:** $\tilde{c}_i = c_i + 0.1$ (eliminate zeros)
- 3 **Graph-diffusion smoothing** ($\gamma = 0.26$):

$$\mathbf{c}_{\text{smooth}} = 0.74 \tilde{\mathbf{c}} + 0.26 P_{\text{adj}} \tilde{\mathbf{c}}$$

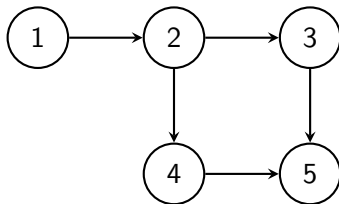
- 4 **L^1 -normalise:**

$$E_N[i] = \frac{c_{\text{smooth},i}}{\sum_j c_{\text{smooth},j}}, \quad \sum_i E_N[i] = 1$$

Result

$E_N[i]$ = probability that a new vehicle appears on link i

Worked Example: 5-Link Network



	Link 1	Link 2	Link 3	Link 4	Link 5
Raw count c_i	0	5	0	3	0
+ Laplace $\tilde{c}_i$	0.10	5.10	0.10	3.10	0.10
Neighbour avg	5.10	1.60	0.10	0.10	—
Smoothed	1.40	4.19	0.10	2.32	0.10
$E_N[i]$	0.172	0.516	0.012	0.286	0.012

Link 1 gets **17.2%** despite zero observations $\Rightarrow$ topology propagates evidence from its busy neighbour

Extending to $2N$: The E_{2N} Vector

Teleportation injects vehicles only into the **Up (active)** phase:

$$E_{2N} = \begin{pmatrix} E_N \\ \mathbf{0}_N \end{pmatrix} \in \mathbb{R}^{2N}, \quad \|E_{2N}\|_1 = 1$$

Physical meaning:

- New vehicles “start” actively driving on a road
- They naturally transition to the Down phase over time
- No vehicles are born directly into local dispersal mode

Combining Topology + Ground Truth

The Power Iteration

$$\mathbf{v}^{(k+1)} = \underbrace{d \cdot M_{2N}^T \cdot \mathbf{v}^{(k)}}_{\text{topology (80\%)}} + \underbrace{(1 - d) \cdot E_{2N}}_{\text{ground truth (20\%)}}$$

$$d = 0.80$$

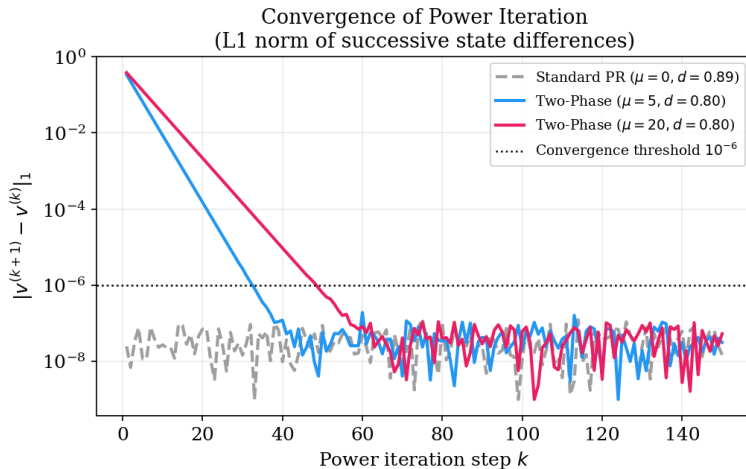
80% of mass follows the
transition matrix
(roads, speeds, physics)

$$1 - d = 0.20$$

20% of mass is re-injected
from **ground truth GPS**
(observed traffic pattern)

Iterate until $\|\mathbf{v}^{(k+1)} - \mathbf{v}^{(k)}\|_1 < 10^{-6}$

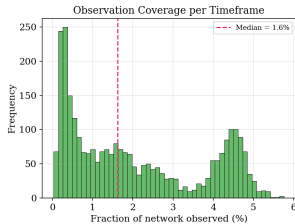
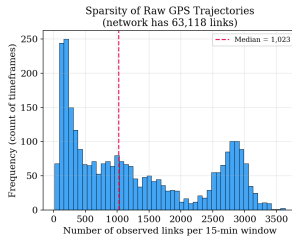
Convergence



Typically converges in ~ 40 – 60 iterations.

Why Smooth? — The Sparsity Problem

- Most links have $c_i = 0$
- Creates:
 - MRE singularities
 - Markov chain dead zones
 - Biased teleportation
- Without smoothing:
MRE = 7,275
- With smoothing:
MRE = 0.19



Graph-Diffusion as a Low-Pass Filter

$$\mathbf{c}_{\text{smooth}} = \underbrace{(1 - \gamma)}_{\text{keep own}} \cdot \tilde{\mathbf{c}} + \underbrace{\gamma}_{\text{borrow from neighbours}} \cdot P_{\text{adj}} \tilde{\mathbf{c}}$$

Spectral view:

Eigenvalue of the filter:

$$h(\lambda) = (1 - \gamma) + \gamma\lambda$$

$$h(1) = 1 \text{ (preserves mean)}$$

$$h(-1) = 1 - 2\gamma \text{ (attenuates noise)}$$

Mass conservation:

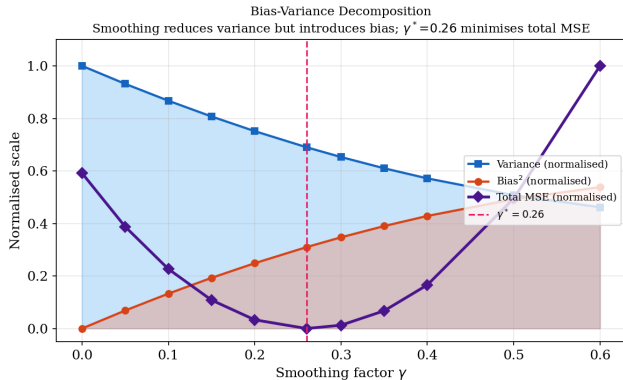
$$\sum_i c_{\text{smooth},i} = \sum_i \tilde{c}_i$$

Smoothing **redistributes**
mass but never creates
or destroys it

The Bias-Variance Tradeoff

$$\text{MSE} = \underbrace{\text{Bias}^2}_{\uparrow \text{ with } \gamma} + \underbrace{\text{Variance}}_{\downarrow \text{ with } \gamma}$$

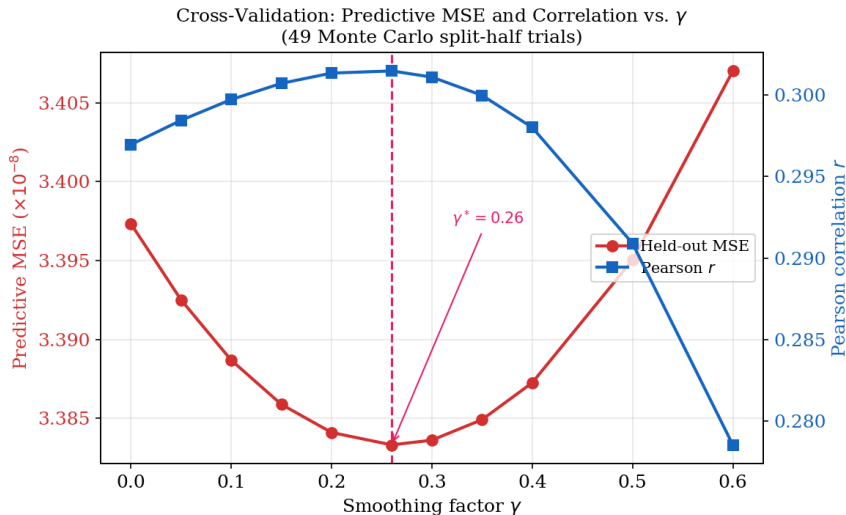
- Too little smoothing → **noisy**
- Too much smoothing → **blurred**
- U-shaped MSE curve
- Minimum **inside** the range



Cross-Validation Protocol (49 Trials)

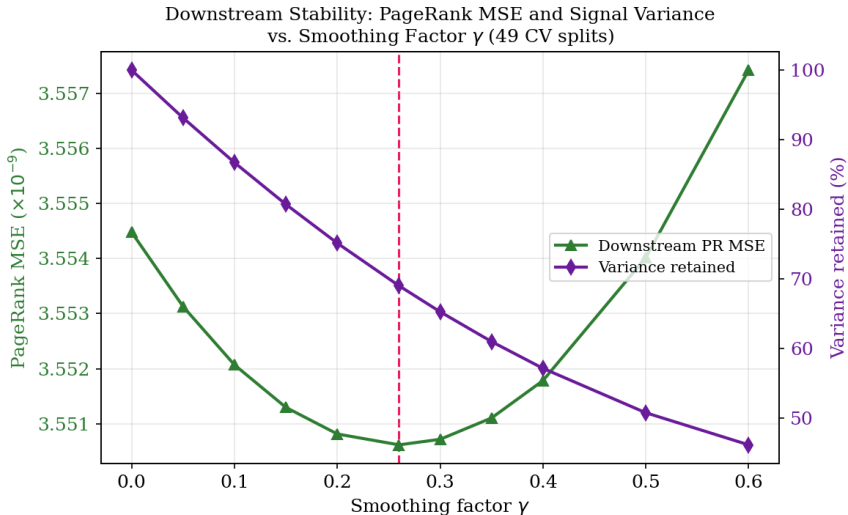
- ① Select 49 timeframe slots (fixed seed = 42)
- ② **Split** GPS data: Set A (train) / Set B (test)
- ③ Smooth Set A with candidate γ
- ④ **Never** smooth Set B (oracle ground truth)
- ⑤ Measure:
 - **Predictive MSE**: smoothed A vs. raw B
 - **PageRank MSE**: PageRank(A) vs. PageRank(B)
 - **Pearson r** : correlation A vs. B
- ⑥ Average over all 49 trials

CV Results: MSE & Correlation



Both metrics optimised at $\gamma^* = 0.26$

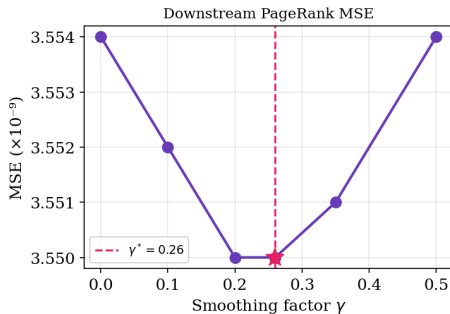
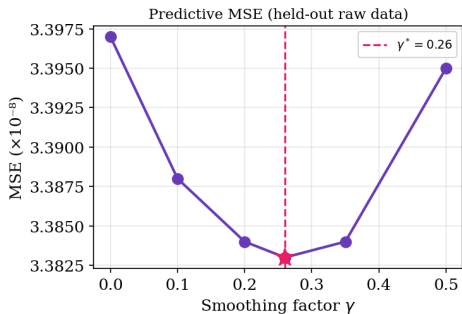
CV Results: Downstream PageRank



PageRank MSE minimum **also** at $\gamma \approx 0.26 \Rightarrow$ input-optimal = output-optimal

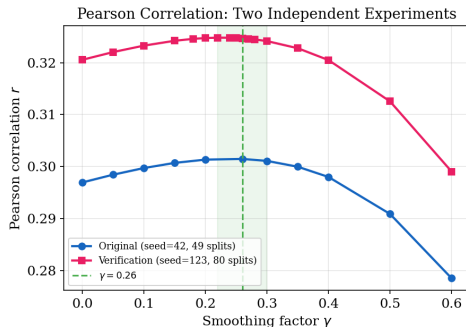
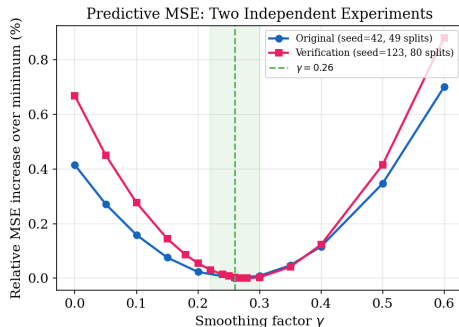
Direct vs. Downstream MSE

Cross-Validation: Optimal Smoothing Parameter ($\gamma^* = 0.26$)



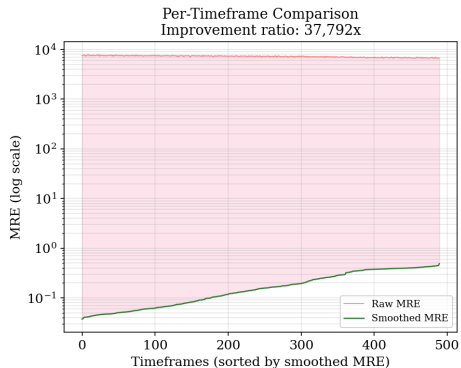
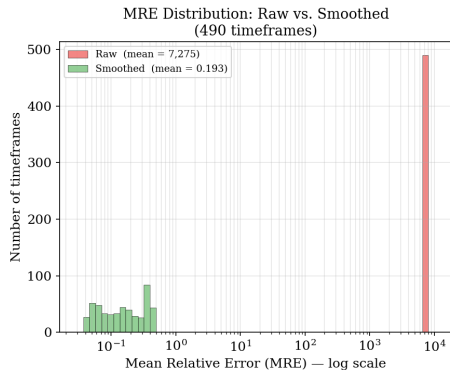
Independent Verification (Seed = 123, 80 Splits)

Independent Verification of Optimal γ^*
(green band = optimal plateau $\gamma \in [0.22, 0.30]$)



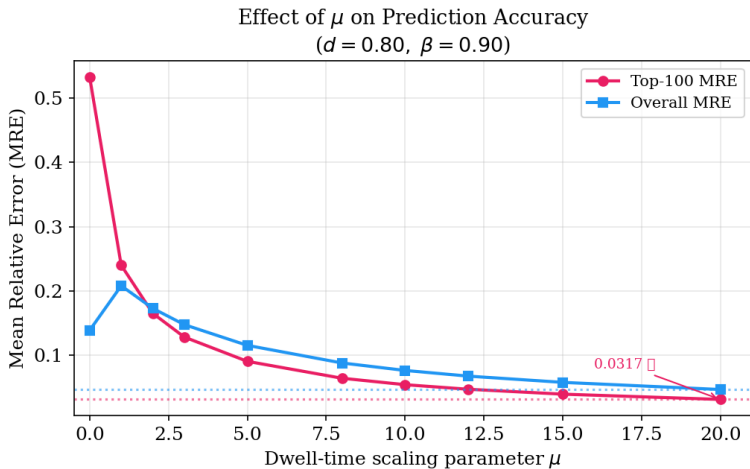
Broad plateau at $\gamma \in [0.22, 0.30]$; $\gamma = 0.26$ is the **centre of the plateau**.
MSE varies $< 0.04\%$ within this band.

Impact of Smoothing: 490-Timeframe Comparison



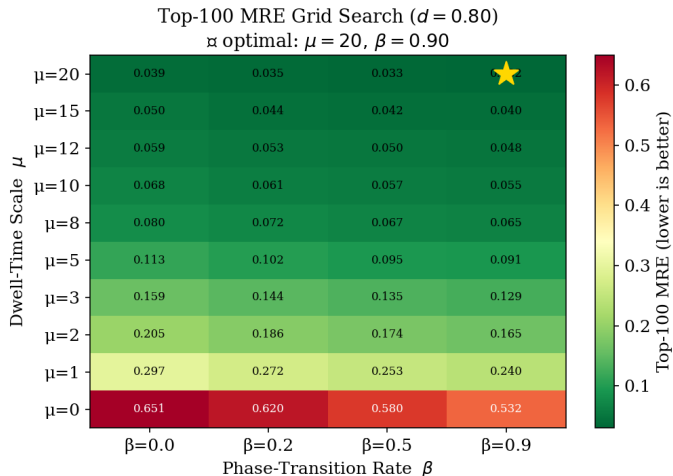
Pipeline	Avg MRE	Improvement
Raw ($\gamma = 0$)	7,275	—
Smoothed ($\gamma = 0.26$)	0.19	$\times 37,800$

Effect of μ (Dwell-Time Scale)



MRE drops monotonically as μ increases. Optimal: $\mu = 20$

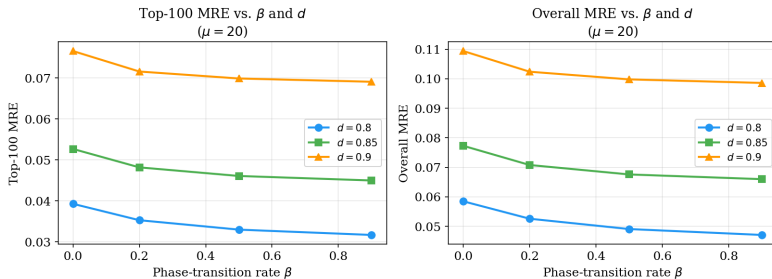
MRE Heatmap: μ vs β



Best: $\mu = 20, \beta = 0.90, d = 0.80$

Damping Sensitivity

Sensitivity Analysis: Damping Factor d and Phase Rate β



$d = 0.80$ balances topology-driven flow with ground-truth injection

The Four Tuneable Parameters

Symbol	Name	Value	Controls
γ	Smoothing	0.26	Evidence propagation strength
μ	Dwell-time	20	Self-loop weight (road physics)
β	Phase-drop	0.90	Active $\rightarrow$ dispersal rate
d	Damping	0.80	Topology vs. ground truth

Transferability

For a new city: get road network + some GPS data $\rightarrow$ grid-search over (γ, μ, β, d)

Key Takeaways

- 1 **Transition matrix** M_{2N} uses **only topology**
(road graph + lengths + speeds)
- 2 **Teleportation vector** E_{2N} uses **only ground truth**
(GPS counts $\rightarrow$ Laplace $\rightarrow$ smoothing $\rightarrow$ normalise)
- 3 **Smoothing is justified** by cross-validation:
 $\gamma^* = 0.26$ minimises MSE (interior minimum, not boundary)
- 4 **Result:** MRE improves from 7,275 $\rightarrow$ 0.19
(37,800 $\times$ improvement with smoothing alone)

Thank You

Questions?